



Interaction between knowledge management, intellectual capital and innovation in higher education institutions

Manuel Alejandro Ibarra-Cisneros¹ · Juan Benito Vela Reyna¹ · Felipe Hernández-Perlines²

Received: 18 May 2022 / Accepted: 28 December 2022 / Published online: 13 January 2023
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023

Abstract

Higher education institutions are essential generators and disseminators of knowledge; however, they must create conditions to lay the foundations supported by knowledge enablers and manage knowledge efficiently. In addition, intellectual capital and innovation are elements that help this process; if everything is correctly articulated, the academic staff and institution will promote better performance. This article analyzes how three knowledge enablers (leadership, culture, and organizational incentives) influence the knowledge management process. In turn, this process encourages the development of better intellectual capital and innovation in the teachers of Higher Education Institutions. A survey applied to 434 academics made it possible to collect information that, through structural equation models of partial least squares, allowed validation of the proposed hypotheses. The study results indicate that organizational culture and leadership significantly influence the knowledge management process. In addition, this process significantly influences intellectual capital and innovation. Finally, the vital link between the variables studied is demonstrated.

Keywords Knowledge Management · Intellectual Capital · Innovation · Enablers · HEIs

1 Introduction

Higher Education Institutions (HEIs) contribute to generating and disseminating knowledge for the benefit of society. In like manner, they are responsible for training high-quality human capital and putting forward projects and research products

✉ Manuel Alejandro Ibarra-Cisneros
manuel_ibarra@uabc.edu.mx

¹ Faculty of Administrative Sciences, Autonomous University of Baja California, Mexicali, Mexico

² Department of Business Administration, University of Castilla-La Mancha, Toledo, Spain

whose clients are firms and the public sector (Cricelli et al., 2018). To reach such goals, knowledge management (KM) and intellectual capital (IC) are fundamental to every institution (Bontis et al., 2007), without leaving innovation (I) aside. Significant contributions to the state of art have been made regarding knowledge, its management, and the benefits for organizations in general (Nonaka & Takeuchi, 1995). Likewise, research work has been carried out in the sphere of HEIs analyzing KM and IC topics, either separately or jointly (De Matos et al., 2020, 2022; Martin-Sardesai & Guthrie, 2018; Quarchioni et al., 2022; Ramírez Córcoles et al., 2011; Rehman & Iqbal, 2020). Which has allowed the opening of a specific research line to study HEIs; such a line helps understand the functioning and interaction of KM, IC, and innovation.

Furthermore, HEIs must be aware that essential parts of their performance are associated with the implementation degree of both KM and IC and I jointly. This way, the better these elements are developed, HEIs have better the opportunities to attract clients, improve relationships with the entire staff and produce new knowledge, innovations, and patents, among other activities (Iacoviello et al., 2019; Secundo et al., 2018). Therefore, it is crucial to understand how the development of KM processes allows for improvement in the acquisition of IC and innovation between the academics of public HEIs. In addition, Iqbal et al. (2019) introduce the analysis of the construct KM enablers into the studied relations to explore which enablers have a more significant influence on KM processes.

One of the objectives of the HEIs authorities is to achieve the excellence of its students and raise the prestige measured through international rankings. To achieve this, it is essential that every institution support and enrich knowledge management processes, as well as strengthen the intellectual capital of its academics. Consequently, the quality and speed of the processes to innovate internally increases. However, like any organization, within the HEIs, some elements may or may not be implemented effectively and are specific and unique to each institution. These elements are called enablers, which are numerous, but this research focuses on three of them, leadership, organizational culture, and incentives.

Empirical evidence has mainly shown that KM, IC, and I individually experience positive relationships. However, their simultaneous behavior and relationship with the three critical enablers have not been thoroughly studied. Based on the above, this research aims to analyze the interactions between KM, IC, and I and the role of the KM enablers of the recognized public HEIs in Mexico. To do so, the following questions are proposed: (i) Does the KM process depend on the correct implementation of KM enablers? (ii) Do KM process, Intellectual Capital, and Innovation impact HEIs performance? (iii) Does IC positively influence the speed and quality of Innovation activities?

To reach the goals, a survey based on the Iqbal et al. (2019) model was applied to 434 academicians from public HEIs accredited at a national level. The hypotheses were verified by employing partial least squares structural equation modeling (PLS-SEM), which was applied to the software Adanco 2.1.1.

This paper contributes to enriching and supporting the empirical evidence developed so far on the critical role KM enablers have so that KM processes are carried out successfully, which contributes to achieving the goals of HEIs. In addition, two other elements are added to the analysis to propose a model that seeks to know how interaction is carried out that benefits academics' teaching, management, and research capacities.

2 Theoretical framework and formulation of hypotheses

2.1 Knowledge management

The knowledge management studies are substantiated by the firm's knowledge-based theory and the resource-based view –RBV–. Such developments aimed to explain how knowledge must be managed to impact the organization, workers, and all the actors it interacts with within its environment. There are well-defined research lines with significant theoretical contributions regarding KM applied in the entrepreneurial sector and education, as in the present case.

First, knowledge is the fundamental factor that individuals and organizations use to generate all sorts of goods and services and satisfy needs; moreover, it is a source of competitiveness. In this sense, knowledge must integrate into the organization via routines, documents, systems, and organizational culture (Fernández-Lopez et al., 2018). Knowledge is classified as explicit and tacit. The former is deemed measurable and is transcribed in books, databases, articles, and all manner of documents; while the latter is acquired through the experience and intuition of each individual; therefore, it is more challenging to communicate to others –in the sphere of HEIs, university staff it is the addition of explicit and tacit knowledge (Ramírez & Gordillo, 2014).

On the other hand, various definitions by KM have enriched state of the art on this topic. Andersen (1999) stated that KM is the addition of all the knowledge of an organization's members, which is systematized and used to improve the organization. Therefore, organizations must be capable of recognizing and administrating the competencies they have (Wen Chong et al., 2000). For their part, Dzenopoljac et al. (2018) defined KM as “the systemic management of all the activities and processes -generation, codification, storage, transference and utilization of knowledge- of the organization and a competitive advantage is attained” (p. 79). According to this definition, KM comprises several stages which, if developed jointly, will allow acquiring a series of advantages for the entire organization to generate productivity, efficiency, cost reduction, and better organization performance (Metha & Tariq, 2020; Davenport, 1994) stated that KM divides into three stages, namely: acquisition, sharing, and use of knowledge. Later, Duhon (1998) divided KM into five stages: identifying, acquiring, assessing, retrieving, and sharing information and, by doing so, generating knowledge. While for Sadeghi Boroujerdi et al. (2019), KM meant acquiring, creating, storing, sharing, and applying knowledge.

Regardless of the differences in the stages, all the definitions concur that to obtain KM, it is at least necessary to acquire, store, share and use knowledge. A series of principles, practices, and techniques are necessary. It is also necessary to measure the support processes and the employees' cognitive capability to identify the benefits produced by KM (Chawla & Joshi, 2011). In the sphere of HEIs, Davenport & Hansen (1999) defined KM as the “organized and systematic process of producing and disseminating information, selecting, distilling and unfolding explicit and tacit knowledge to create a single value that can be utilized to strengthen the teaching-learning environment” (p. 99).

HEIs use their KM to foster a competitive advantage (Barley et al., 2018; Mubarak & Sabraz Nawaz, 2019) over the rest of the institutions. Moreover, KM allows for optimizing the knowledge generated and operating efficiently in the knowledge society. Davenport and Hansen (1999) argued that for HEIs to maximize their KM, they must have a culture that promotes knowledge creation. They also have to acknowledge their strengths, use information technologies correctly, keep good relationships with the environment, generate and share new knowledge, be open to changes and resort to KM techniques.

The first KM stage is the *acquisition*, which comprises searching, identifying, gathering, and mapping knowledge (Pinho et al., 2012), which can be obtained from internal or external sources (Obeidat et al., 2016). In the case of HEIs, these sources are the government, the entrepreneurial sector, graduates, NGOs, other HEIs, and society. It has to be underscored that this is not a process that can be planned or controlled. Moreover, it may be acquired or created by the organization itself (Dzenopoljac et al., 2018); this way, the firms that invest in R&D produce their knowledge. However, this process is expensive and complicated to develop, and because of this, not all HEIs can undertake it efficiently. *Codification and storage* consist in codifying and classifying knowledge and storing it to be utilized by the right people (Dzenopoljac et al., 2018). Knowledge is stored in databases, documents, processes, and systems. *Knowledge Sharing* (KS) is the process of disseminating knowledge among all the members of an organization. KS turns organizational knowledge into individual or group knowledge when internalization and socialization occur (Wang et al., 2014). For its part, knowledge *use* is applied to the organization's functions to carry out activities that may have direct results, such as products, services, procedures, and regulations, as well as generate value for the organizations (Dzenopoljac et al., 2018).

2.2 KM enablers

According to Rivera and Rivera (2016), KM enablers are defined as “the set of organizational, cultural and diverse elements that exist in the organization which supports and facilitate the processes of creation, storage, transference and application of knowledge” (p. 24). This is to say, they are elements that, if correctly applied, allow KM to unfold and not remain only as acquired knowledge; therefore, enablers contribute to the growth of the km process (Sahibzada et al., 2020a, b).

There are various enablers that organizations utilize; among the most distinguished there are organizational culture, leadership, and incentives of the organization.

As regards *organizational culture* (OC), it is supported by the organization's mission and vision (Wen Chong et al., 2000), and also by the way the processes and managerial practices are devised and carried out; as well as the attitudes and values of the organization (Adeinat & Abdulfatah, 2019), the beliefs that compose their identity such as performance criteria, assessment of the staff and motivation (Mubarak & Sabraz Nawaz, 2019). Likewise, it is essential to mention that the organizational culture encourages knowledge creation (Thani & Mirkamali, 2018). In the case of HEIs, the problem with organizational culture is that members consider knowledge as their individual property; thereby, they do not want to share it with the rest of the institution or external parties (Mubarak & Sabraz Nawaz, 2019). *Organizational incentives* (OI) are rewards received by the members of an organization owing to their activities, either bonuses, recognitions, or awards, among others. Incentives make sharing knowledge among organization members easy and allow reinforcing such behavior (Cho & Korte, 2014). In the education sphere, incentives promote collaborative work between academics as it is a way to obtain benefits –promotions, prestige– (Cheng et al., 2009); HEIs members are further motivated by personal incentives (Bercovitz & Feldmann, 2006). For its part, *leadership* (LS) “implies a leader's capability to align KM behaviors with organizational strategies, promote KM values, communicate the best strategies and provide assessment metrics to measure the impacts of knowledge” (Mubarak & Sabraz Nawaz, 2019, p. 192). In the education sector, leadership enables academics to help reach new goals utilizing applying organizational strategies. In this sense, Shariq et al. (2019) agree with leadership's importance in achieving KM processes' development. Finally, it is essential to highlight that organizational culture and leadership are two of the most used facilitators in the academic literature.

2.3 Intellectual capital

IC is composed of all an organization's intangible assets, including relations between all the organizational resources (Dias Jordão & Casas Novas, 2017). In the case of HEIs, Ramírez and Gordillo (2014) stated that IC refers to the institutions' non-tangible or non-physical assets such as the processes, innovation capacity, patents, their members' tacit knowledge, abilities, talents, society recognitions, networks of collaborators, among others. These indicators are directly linked with HEI's development and growth (Pedro et al., 2019). According to Cricelli et al. (2018), in an institution, the professors' and researchers' IC comprises the abilities and competencies to teach and research; in point of fact, there was a positive relationship between the docent's IC and the students' labor training (Sultanova et al., 2018). Likewise, they expressed that administrative staff transforms tacit knowledge into explicit and integrates it into the institution's structure, while students take the professors' knowledge to society and vice versa. Furthermore, it is worth underscoring that IC in universities varies in function of their proper profile; this is

to say, they are oriented to research, teaching, or entrepreneurship (Secundo et al., 2016).

IC is usually divided into three dimensions –human capital (HC), structural capital (SC), and relational capital (RC)–. *Human capital* is the set of explicit and tacit knowledge of the HEIs staff acquired through formal and informal education processes (Bontis et al., 2007). It is a collective capacity and capability to extract the best outcomes from the knowledge of the individuals (Qurashi et al., 2020). Following Secundo et al. (2016), HC is composed of the “intangibles found in individual competencies such as the expertise and knowledge of the professors, researchers, students and in general of everyone working for an institution” (p. 422). HC becomes especially relevant as it carries out the essential teaching, researching, and management duties; its responsibility is to train excellent human resources, the reason why its competencies must be ascertained to increase them (Demuner et al., 2016). Furthermore, HC is strongly associated with KM (Supapawawisit et al., 2018).

For its part, *structural capital* is defined as explicit knowledge related to the internal process of dissemination, communication, and management of scientific and technical knowledge in the organization (Bontis et al., 2007; Booker et al., 2008; De Frutos-Belizón et al., 2019; Leitner et al., 2014; Secundo et al., 2016) pointed out that SC “comprises the resources in the organization itself, namely: databases, procedures, manuals, intellectual property, research projects and infrastructure, processes and routines, values, ethics of the organization members, university culture and team philosophy” (p. 422). It is also necessary to include scientific and technological knowledge management, bibliographic repositories, and internal dissemination processes (Ramírez & Gordillo, 2014).

Relational capital is the set of economic, political, and institutional relations developed and maintained by universities (Leitner et al., 2014). RC incorporates the lessons from its members’ internal interactions and the links with the socioeconomic environment producing a benefit for both parties (Demuner et al., 2016). Therefore, RC contributes to disseminating knowledge gained by professors, researchers, and students (Paoloni et al., 2019). As regards the relations kept by universities, these are maintained with entrepreneurs, governments, non-governmental organizations, business chambers, graduates, and various authorities (Ramírez et al., 2012), which is vital to be made known online, so that the actions of the HEIs have a greater reach to all stakeholders (Ramírez & Tejada, 2019).

2.4 Innovation

Innovation may be understood as introducing improvements in the organizations’ processes (Damanpour et al., 2009); it also comprises all the procedures, processes, practices, technologies, and services set up to meet the client’s demands (Rezaei & Moghanlo, 2015). Every organization has the opportunity to innovate, though this will depend on its internal capabilities, technologies, and know-how to develop its knowledge (Tether, 2003). This way, the organizations that innovate are those that convince themselves and effectively harness their abilities and strengths. In the education sector, innovation takes place through the improvement of its processes, the

generation of new competencies taught to the students, and responsiveness to face the contingencies and challenges that may appear.

Innovation speed (IS) is defined by Iqbal et al. (2019) as the capability to introduce new academic programs, learning methods, and know-how to face challenges in a complex and turbulent environment. In this regard, it is necessary to consider that innovative HEIs are pioneers in developing science, culture, academia, and management and are followed by other HEIs. On the other hand, *innovation quality* (IQ) is defined as the ability to offer innovative educational services that are better than the competitors' (Iqbal et al., 2019). In this sense, quality must be supplemented with speed, as it is the first and better in offering educational, management, and research services.

Raj Adhikar (2010) argued that in the current dynamical world, successful HEIs are those who continuously produce new knowledge and widely disseminate it over their systems, and this takes place because of their high position in innovation at national and international levels. Therefore, it is essential to emphasize that innovation in HEIs must be part of their DNA; it has to be in their organizational culture, and all the professors, researchers, administrative staff, and students must participate in it (Lin, 2007).

2.5 KM enablers and KM process

KM enablers are critical elements within organizations so that knowledge-related activities are carried out efficiently (Ferasso & Grenier, 2021). The KM process comprises different KM stages; KM processes are defined as the activities carried out so that the members of an organization can produce, store, share and use knowledge (Rivera & Rivera, 2016; Sahibzada et al., 2022). These activities are addressed through internal or external processes (Torres et al., 2018).

Empirical evidence has been shown (Cho & Korte, 2014; Iqbal, 2021; Iqbal et al., 2019; Obeidat et al., 2016; Raudeliūnienė & Kordab, 2019; Sahibzada et al., 2020a, b; Veer Ramjeawon & Rowley, 2020) how KM enablers facilitate the performance of the KM process. By these enablers, each KM stage will proceed correctly and in the correct order. In the case of HEIs, the optimal development of KM processes is carried out through organizational culture, innovation, and leadership, which allow academics to be motivated to develop KM within a culture oriented towards generating and transmitting knowledge.

According to Donate and Sánchez de Pablo (2015), leadership allows the development of channels and initiatives to carry out the best KM practices, so leadership style, motivation, and communication are fundamental. Likewise, intelligent leadership increases the interaction opportunities of the members of an organization (Ding et al., 2019).

As Haider et al. (2022) pointed out, in higher education institutions, leadership is transcendental for Department Heads and Deans since responsible leadership fosters communication and constructive relationships between staff. In addition, a positive social climate is generated, and knowledge sharing is inspired. Therefore, to the extent that there is positive leadership in HEIs, the process of managing knowledge

among all institution members is facilitated, which results in better conditions to innovate, learn, interact with the environment, and increase the productivity of staff and the institution at a global level.

For its part, culture is a critical enabler that allows knowledge to be shared within the organization (Al-Kurdi et al., 2018). Therefore, if there is a culture that promotes knowledge sharing among academics, it will be possible for the KM process to be carried out optimally. Thus achieving and encouraging innovation and productivity by all academics. However, it is essential to note that the cultural context varies from country to country. Therefore, expecting the same behavior within HEIs in different countries is impossible. This is shown by international evidence (Ardichvili et al., 2006).

However, it is also important to mention that “the competitive nature of academia and the need for an ongoing publication to be considered employable are factors inhibiting the sharing of knowledge among academics within HEIs” (Al-Kurdi et al., 2018, p. 234). Therefore, it is necessary to investigate if the conditions exist for KM to be developed.

In this manner, the following research hypotheses were proposed:

H1: KM enabler organizational culture positively influences the KM process.

H2: KM enabler organizational incentive positively influences KM process.

H3: KM enabler leadership positively influences the KM process.

2.6 KM process and innovation

Innovation cannot generate itself; knowledge must be acquired, shared, and used so that it is transformed and later disseminated in new educational programs or administrative and academic processes. This way, the KM process influences the innovation of processes in HEIs. To accomplish this, it is necessary that the KM process articulates with the institution members, and they introduce improvements inside it to later work with the entrepreneurial sector. Evidence positively relating the KM process and organizational innovation has been found (Agostini et al., 2017; Ling & Nasurdin, 2010; Teixeira et al., 2018). Likewise, Ling and Nasurdin (2010) pointed out that the dimensions of knowledge management are considered the core driving power of administrative innovation.

This research is intended to verify the generated evidence but focuses on the education sector, where KM processes influence both the quality and speed of organizational innovations. This means that better implementation of KM processes results in higher quality and speed innovation. Innovation in the HEIs translate into a better educational offer, better services for its clients, new lines of research, and new ways of satisfying needs and contributing to science, technology, and society. This will make it possible to attract more clients, investigate contracts, improve society’s reputation, and be better positioned in the international rankings of universities. Thus, hypothesis four is stated:

H4: KM process positively influences innovation.

2.7 KM process and intellectual capital

The evidence produced by studies (Hayaeian et al., 2022; Pflugfelder, 2021; Wang et al., 2016) verified the existence of a close relationship between KM and IC. According to Dias Jordão and Casas Novas (2017), KM was the drive for the growth and development of IC. Additionally, KM enables organizations to define strategies to produce IC. That is to say, for HEIs to apply IC efficiently, it is necessary first to develop KM processes, which for their part, are preceded by KM enablers. This statement is supported by Cabrilo and Dahms (2018) by indicating that the configuration of IC in organizations must be well understood before setting up KM processes. Analyzing the above, the hypothesis is:

H5: KM process positively impacts intellectual capital activities.

2.8 Intellectual capital and innovation

In recent years, significant evidence has been generated that supports a close relationship between IC and product, service, and organizational innovation. (Akil et al., 2021; Allameh, 2018; Alshamsi et al., 2019; Wang et al., 2021; Waseem et al., 2018; Wendra et al., 2019). As Allameh (2018) pointed out, innovation requires powerful stimuli that can produce innovative results by putting the right people together with the right intellectual culture. In this way, innovation is the product of well-prepared and consolidated intellectual capital; therefore, as the IC is encouraged by the HEIs, innovation will increase, improving educational quality, positioning universities on a national and global scale, and allowing society to meet expectations. Therefore, it is proposed as a hypothesis:

H6: There is a positive and significant relationship between Intellectual capital and HEIs Innovation.

3 Methodology

3.1 Sample and data collection

The population under study was the full-time professors and researchers in public higher education institutes recognized by education authorities. According to Secretaría de Educación Pública, [Secretariat of Public Education] (2018), there are 7,442 universities in Mexico, where about 4000,000 professors work; but only 94,717 full-time professors were the objects of study. Therefore, a sample of 434 surveys to apply was decided with a margin of error of 4.2% and a confidence level of 95%.

The sample consisted of 52% men and 48% women, the average age of the professors was 42 years, 52% had a doctorate, and the rest had a master's degree. Likewise, 86% obtained their last degree of studies in a national institution and the rest abroad, 37% have a seniority of up to 10 years, another equal percentage between 11 and 20

Table 1 Pearson's correlation indices between the variables of the model

Construct	KM enablers	KM process	IC	Innovation
KM Enablers	1	0.877**	0.805**	0.697**
KM Process	0.877**	1	0.865**	0.744**
IC	0.805**	0.865**	1	0.809**
Innovation	0.697**	0.744**	0.809**	1

**, Correlation is significant at the 0.01 level (2-tailed)

Table 2 Powers reached in each correlation of the model

Construct	Correlation ρ H1	A err prob	Sample size achieved	Power (1- β err prob)
KM Process ---> KM Enablers	0.877**	0.02	434	1
KM Process ---> IC	0.865**	0.02	434	1
KM Process ---> Innovation	0.744**	0.02	434	1
IC ---> Innovation	0.809**	0.02	434	1

years, and 8% are over 30 years of antiquity. On average, professors taught 12 h of class per week at the undergraduate or graduate level; 86% were full-time professors, and the rest were researchers.

To verify that the achieved sample of 434 teachers surveyed complies with the reliability parameters that justify its analysis, the power level achieved through the Gpower program (Faul et al., 2009) was verified. For this, as a first step, Pearson's correlation values were determined between the KM process and KM enablers (KMp-KMe), reaching an index of 0.8.77. The second correlation generated was the KM process, with CI obtaining 0.805, the third correlation between the KM process and innovation with 0.744, and the correlation between IC and Innovation reached 0.809 (Table 1).

As a next step, the power was calculated with the above data, as presented in the previous table. With the help of the G*power software, the established criteria were a bivariate correlation test of a standard two-tailed model and considering a type of analysis in post hoc mode, having already applied 434 surveys (Table 2). Finally, with a demanding value of $\alpha=0.02$ (2%), which is equivalent to being wrong 2 out of 100 opportunities for each of the cases, a power of 100% confidence was obtained in each of the analyzed correlations, clearly above > 0.80 minimum acceptable (Kline, 2015). This means that the applied sample is reliable for further analysis.

The survey was provided by (Iqbal et al., 2019), who researched the relationship between the same variables, but in Pakistan, which was built from contributions by (Huang & Li, 2009; Wang & Wang, 2012). In addition, the instrument was adapted for the case of Mexico, with a total of 71 items; the –original survey has 62 items-, divided into five sections.

The data-gathering process consisted of the identification of all the accredited Mexican HEIs; then, the websites of such institutions were visited in order to

collect the most significant possible number of email addresses of professors that engage in teaching, management, and research. Finally, the instrument was applied using an internet survey; after sending more than 1,500 emails, in the end, 434 were validated.

In order to verify that the use of internet surveys did not generate common biases that could have distorted the results. Harman's test was performed to check if any factor largely explains the variation in the data. In this case, the result was 46.37, which indicates that the data is adequately associated with the different constructs analyzed (Tehseen et al., 2017).

It is essential to point out that Mexican universities have three profiles of professors. The full-time oriented mainly to teaching; those of full time oriented mainly to investigation (they are accredited before the National System of Researchers by the National Council for Science and Technology. That is to say; they are professors who, in addition to carrying out all the activities of a professor with teaching-oriented and administrative management tasks, have official research tasks of 20 h per week with complementary remuneration); and partial time subject professors. The latter are professionals with a high level of experience in the private or public sector who are hired to teach certain subjects. These professors are valuable since they transmit what the productive sector is doing in practice. Therefore, their experience allows them to have another vision of the methodological or theoretical aspects taught by professors and researchers. However, for this research, they could not be considered because there are no public records to access to interview them; although some universities may have information about their contribution, there is no data that this research has been able to access.

Therefore, it was decided to consider only full-time professors, who mainly carry out academic activities (teaching classes as a priority, although they also carry out consulting, tutoring, design of educational plans, accreditations, and research, among other inherent activities); or the research-oriented professor as the main profile. Without neglecting the other activities mentioned, this type of professor has priorities in a different hierarchical order than the one oriented to the academy (publication of articles in prestigious international and national journals, patents, funded research projects, design of public policies, sectoral, thesis directions, teaching classes, scientific popularization mainly).

3.2 Measurement

The model proposed is composed of four constructs (Fig. 1). **First, KM enablers** were divided into three dimensions (first-order construct): *Leadership* (LS) with five items, which measure the actions of KM fostered by school authorities –for example, there is an established vision for KM in the institution; The Institutional Development plan focuses on KM; the authority of the institution recognizes that knowledge assets can add value; the authority of the institution shows commitment to initiatives to promote KM-. *Organizational culture* (OC) with five items that measured how OC enables developing actions aimed to improve KM –for example, communication of success stories is widely encouraged within the university; knowledge does not threaten university positions; the

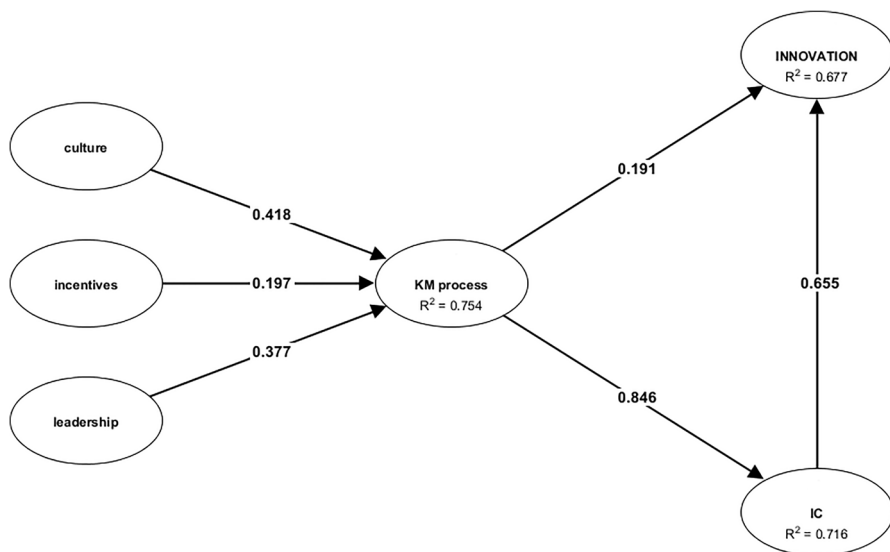


Fig. 1 Proposed model

university aligns rewards and recognition with knowledge activities; there is a culture of openness and trust; the university encourages empowerment and participation in decision-making-. *Organizational incentives* (OI) with five items that analyzed if HEIs set up systems in order to support professors to accomplish the organization's goals –for example, the acquisition of knowledge, the exchange and the use of these are rewarded with a higher salary; or a higher bonus; or with a promotion-.

The **KM process** is a second-order construct that comprises three dimensions: *knowledge acquisition* (KA) with six items to find out if HEIs have defined mechanisms to acquire the knowledge which later will be used –For example, the institution offers an open environment for scholars to acquire new knowledge; the institution actively observes and adopts best practices in the education sector; Information relevant to our activities is continuously collected; knowledge is obtained from different sources: academics, business people and society-. *Knowledge Sharing* (KS) with six items that measured the existence of mechanisms to ensure that knowledge is shared by actors both inside and outside the institution, -example: information and knowledge necessary for the tasks are shared academic; knowledge is exchanged between academics and authorities to achieve goals with little time and effort; information systems such as intranet and electronic bulletin boards are developed to share information and knowledge; knowledge is shared between managers and academics-. *Knowledge use* (KU) with six items as well, which measured if HEIs use knowledge to improve their services, -example: the university effectively manages different sources and types of knowledge; the university uses the available knowledge to improve the services provided to its clients; the university applies the available knowledge to improve its performance-.

Intellectual capital is a second-order construct composed of 21 items divided into three dimensions; the first one is *HC*, which analyzes if HEIs incentivize the development of competencies and abilities of professors and also boost creativity and exchange of ideas (Secundo et al., 2016), -example: employees have adequate work experience to carry out their work successfully in our university; the university offers well-designed training programs; employees of our university often develop new ideas and knowledge-. For its part, *SC* identified the tools to manage the information in the institution.- For example, our university responds to changes very quickly; our university has an easily accessible information system; our university's systems and procedures support innovation-. *RC* allowed finding out which external actors HEIs have relations with -for example, our university maintains appropriate interactions with its interest groups; or long-term relationships with the community; our university has many excellent providers-.

The construct **Innovation** (second order) was divided into *Innovation Speed* (IS) and *Quality* (IQ), both with five items. The first measured the speed of HEIs to devise and launch new programs, solve problems and generate improvement processes (Iqbal et al., 2019). The second, for its part, was focused on finding out if academicians deem innovations in their HEIs better than those of competence (Rezaei & Moghanlo, 2015).

Response type. All items were designed as 5 points on a Likert scale, where 1 = strongly disagree and 5 = strongly agree. Therefore, they were responses on an ordinal scale that allowed the respondent to have greater freedom to generate his response based on his perception of his actions in his institution.

4 Results

The results were analyzed in two stages: the measuring and structural models.

4.1 Measuring model

In this second stage, partial least squares structural equation modeling (PLS-SEM) employing statistical software ADANCO 2.1.1 (Henseler & Dijkstra, 2015). This technique was considered the most suitable owing to the use of non-normal data in structures of complex models (Ringle et al., 2018).

The results of the measuring model indicated that all the analysis constructs were reliable as they have values over the minimum established (Table 3). For example, in the case of Dijkstra-Henseler's (ρ_A), all the constructs have values over 0.700 (Henseler et al., 2015); and the same case for composite reliability (ρ_c) (Fornell & Larcker, 1981), as well as for Cronbach's alpha (Hair et al., 2014).

The analysis of factorial loads indicates that the items belonging to each construct must have a minimum load of 0.707 to be considered part of such constructs (Carmines & Zeller, 1979). In this stage, 36 items met this parameter (Table 4).

Table 3 Construct reliability

Construct	Mean	SD	(ρ_A)	(ρ_c)	(α)
KME OC	3.40	0.90	0.848	0.929	0.848
KME OI	3.06	1.15	0.906	0.934	0.906
KME LS	3.63	1.00	0.925	0.943	0.924
KM process	3.40	0.96	0.944	0.953	0.944
Intellectual capital	3.42	0.88	0.977	0.980	0.977
Innovation	3.22	1.06	0.937	0.949	0.937

Another indicator analyzed was Average Variance Extracted (AVE), which defines the variation of a construct regarding its items; in any case, such value must be superior to 0.50, according to Fornell and Larcker (1981). In this case, all the constructs were located over 0.700 (Table 5). In the same table, discriminant validity was also analyzed, which states that a construct shares more variation with its indicators than another (Hair et al., 2014). In this case, it is noticed that all the constructs met this indication since the values in the diagonal of the matrix are above the values before them in each column.

Another discriminant validity measure, the Heterotrait-Monotrait Ratio of Correlations (HTMT), was included in the analysis. This indicator defines that when a model is correctly fitted, its values must be under 0.90 (Gold et al., 2001). Table 6 shows that all the values were below this parameter; thereby, it is considered that the model fit is correct.

4.2 Structural model analysis

The third stage to obtain results consisted in generating the structural model, for which a collinearity analysis of the constructs was run as a first approximation to the final results. Again, the non-existence of collinearity was demonstrated, as the values ranged from 1 to 10.

The following step was to ascertain the determination coefficients (R^2) of the dimensions KM enablers, KM process, IC, and I. These dimensions resulted from previously analyzed constructs; a type A second-order model A analysis was utilized to verify the research hypotheses.

All the constructs had a significant R^2 : KM process (0.754), Innovation (0.677), and Intellectual Capital (0.716). According to Chin (1998), the predictability levels of these coefficients were substantial for the KM process, IC, and I. This way, it is considered that all the constructs have significant explanatory levels.

Finally, the direct effects of the analyzed dimensions were ascertained using the same system to verify the research hypotheses. The t-values, p-values, and path coefficients (β) were calculated using bootstrap resampling for 4,999 subsamples (Table 7). It is noticed that all the relations presented positive values. The relations between KME organizational culture and KM process ($\beta=0.418$, $t=8.584$, sig. = 0.000, $p<0.01$); KME organizational incentives and KM process ($\beta=0.197$,

Table 4 Factor loadings

Indicator	LS	OC	OI	KM process	IC	Innovation
LS1	0.8821					
LS2	0.8984					
LS3	0.8803					
LS4	0.8973					
LS5	0.8231					
OC4		0.9317				
OC5		0.9321				
OI1			0.8929			
OI2			0.9046			
OI3			0.8829			
OI4			0.8527			
KA4				0.8183		
KS1				0.8532		
KS2				0.8584		
KS4				0.8430		
KS5				0.8473		
KU2				0.8401		
KU3				0.8627		
KU4				0.8585		
SC2					0.8381	
SC4					0.8392	
SC6					0.8654	
SC7					0.8519	
RC1					0.8787	
RC2					0.8534	
RC5					0.8404	
IS1						0.8744
IS2						0.9011
IS3						0.9272
IS4						0.9175
IS5						0.9217
IQ1						0.9194
IQ2						0.9095
IQ3						0.9175
IQ4						0.9243
IQ5						0.9114

$t=5.561$, sig. = 0.000, $p < 0.01$); and KME leadership and KM process ($\beta=0.377$, $t=8.864$, sig. = 0.000, $p < 0.01$) presented a significant association that allows confirming *H1*, *H2* and *H3*. For its part, the association between KM process and innovation ($\beta=0.191$, $t=3.310$, sig. = 0.000, $p < 0.01$) was also considered significant

Table 5 Convergent and discriminant validity Fornell-Larcker criterion

Construct	AVE	OC	OI	LS	KM process	INNOVATION	IC
KME OC	0.8685	0.8685					
KME OI	0.7805	0.3448	0.7805				
KME LS	0.7685	0.4791	0.3161	0.7685			
KM process	0.7187	0.6313	0.4280	0.6037	0.719		
INNOVATION	0.8327	0.3708	0.2917	0.4086	0.5557	0.8327	
IC	0.7269	0.5348	0.3654	0.4903	0.7162	0.6671	0.7269

Table 6 Discriminant validity: Heterotrait-Monotrait Ratio of Correlations (HTMT)

Construct	OC	OI	LS	KM process	INNOVATION	IC
KME OC						
KME OI	0.6696					
KME LS	0.7802	0.6135				
KM process	0.8881	0.7065	0.8300			
INNOVATION	0.6683	0.5730	0.6717	0.7743		
IC	0.8198	0.6552	0.7514	0.8979	0.8523	

Table 7 Model results (direct effects)

Hypothesis	Effect	(β)	Se	<i>t</i> -value	<i>p</i> -value	Cohen's f^2
H1	KME organizational culture -> KM process	0.418	0.0487	8.584	0.000***	0.3292
H2	KME organizational incentive -> KM process	0.197	0.0354	5.561	0.000***	0.0958
H3	KME leadership-> KM process	0.377	0.0425	8.864	0.000***	0.2793
H4	KM process -> Innovation	0.191	0.0577	3.310	0.000***	0.0322
H5	KM process -> Intellectual Capital	0.846	0.0156	54.298	0.000***	2.5237
H6	Intellectual Capital -> Innovation	0.654	0.0540	12.129	0.000***	0.3774

***: $p < 0.01$

but modest, even so, the statistics are positives, hence H4 is accepted. The relationship between the KM process and IC ($\beta = 0.846$, $t = 54.29$, $\text{sig.} = 0.000$, $p < 0.01$) was the highest of all the relations analyzed; therefore, H5 is also accepted. The relation between IC and Innovation ($\beta = 0.654$, $t = 12.12$, $\text{sig.} = 0.000$, $p < 0.01$) presented a high and significant association, therefore H6 is accepted.

Finally, the size effects (Cohen f^2); in two of the relations, there are large, other two moderate, and two small, according to the parameters established by Cohen (1988) –small-effect value: $0.02 \leq$ and ≤ 0.15 ; moderate effects $0.15 \leq$ and ≤ 0.35 ; while large effects ≥ 0.35 .

4.3 Discussion and conclusions

The results showed that all interactions between the constructs are favorable; this is to say, all the hypotheses are accepted. However, it is necessary to analyze such relations' behavior thoroughly. The first hypothesis is *H1: KM enabler organizational culture positively influences the KM process*. The results agreed with Adeinat and Abdulfatah (2019) and Ferasso and Grenier (2021) regarding the importance of organizational culture in contributing to better knowledge management by professors. A culture that stimulates the generation and sharing of knowledge promotes a healthy and stimulating environment to improve the various processes and objectives of HEIs. The second hypothesis: *H2: KM enabler organizational incentive positively influences KM process*. Although the result is similar to that obtained by Iqbal et al. (2019), being positive, its small effect on the KM process translates into a low influence of the incentives towards a better KM process. An explanation for this result suggests that teachers do not carry out their academic activities based on their perceived incentives. On the contrary, the culture of generating and sharing knowledge with students, organizations, and society is the primary source of motivation. This is derived from the intrinsic commitment of professors to contribute to creating a better society. For its part, *H3: KM enabler leadership positively influences the KM process*. The evidence supports the results of Raudeliūnienė and Kordab (2019) and Sahibzada et al. (2020a, b). In this regard, adequate leadership by the authorities of the institutions facilitates the establishment of policies aimed at efficiently managing knowledge. Furthermore, a good leader makes it possible to support those innovative projects that improve the performance indicators of the institutions, which subsequently leads to greater recognition and prestige of both the institution and the professors themselves.

The results support the importance of KM enablers for the KM process's adequate performance. In this sense, the incentives granted to academics, the organization's culture, and leadership promote better knowledge management of HEIs. Likewise, the results obtained by Obeidat et al. (2016) were confirmed. Analyzing each of the three selected enablers, it was found that the organization's culture and leadership significantly influence knowledge management processes. *H4: The KM process positively influences innovation*; it agrees with Damanpour et al. (2009) and Iqbal et al. (2019), as there is a positive relationship between the KM process and innovation. However, the degree of influence is marginal, so although the results are positive, it is crucial to be cautious.

It is essential to mention that the HEI's innovation processes directly result from the knowledge captured, processed, and shared. Therefore, if this knowledge is better managed, it will result in better innovation processes. Likewise, organizational innovation can be increased through leadership and the strengthening of KM capabilities (Ding et al., 2019).

For its part, *H5: KM process positively impacts intellectual capital activities*, the findings of various authors (Hayaeian et al., 2022; Pflugfelder, 2021) on how the KM process influences IC is verified. In this respect, the results corroborate such studies since there are strong correlations between variables. Moreover, both variables have been extensively analyzed, and the results corroborate the

importance of KM so that IC activities are developed well. Finally, *H6: There is a positive and significant relationship between Intellectual capital and HEIs Innovation*. It agrees with the results (Alshamsi et al., 2019; Wang et al., 2021; Wendra et al., 2019) in that the IC encourages innovation activities, in this case, among teachers of the HEIs. In this regard, a qualified and motivated human capital; a well-planned and organized structural capital with clear administrative policies; and a relational capital based on frequent contact with graduates, entrepreneurs, communities, governments, and the rest of society, favor both the quality and the speed of the innovation processes are developed more quickly and efficiently. Therefore, academic and research activities at HEIs provide results that translate into patents, books, scientific articles, and consultancies to other organizations.

It is essential to mention that KM practices can vary depending on the type of sector (public or private), as well as the sociodemographic context (Pepple et al., 2022); Therefore, these results cannot necessarily be extrapolated to the rest of the HEIs in the rest of the world, where the activities of the teachers tend to vary.

It is concluded that HEIs need to develop KM processes from KM enablers, especially the organizational culture and leadership of all central authorities and institutes. With this, it is ensured that the knowledge is disseminated among all the members of the HEIs, and later, said knowledge reaches the entire society. Likewise, it is necessary to promote knowledge exchange between academics, administrative staff, management, and students to avoid resistance to change and individualism. At the same time, HEIs must identify their intellectual capital and develop it based on KM processes, where rapid and quality innovation is the result of both variables.

Finally, the prestige of an institution is the result of the excellent use of knowledge management, intellectual capital, and innovation. Therefore, the contribution of all teachers is essential for achieving the objectives of all HEIs, regardless of their status, profile, or employment status.

4.4 Contribution

This article presents the following contributions to KM's state of the art in HEIs. Firstly, it was demonstrated that enablers are the cornerstone of good KM-process performance. In the literature, this relation is obviated; nevertheless, it is not defined which enablers are more defining in the development of KM. Furthermore, out of the three analyzed enablers, organizational culture had the most weight, while incentives were the least. Additionally, it was shown that knowledge management positively influences a better development of both intellectual capital and innovation. Likewise, intellectual capital also contributes by improving the speed and quality of the innovation developed by full-time professors, regardless of the profile of each one of them.

4.5 Limitations

The study's main limitation is that it was only applied to full-time academics and not to senior managers and decision-makers, who may have a different perception of the topic. Unfortunately, the lack of information and databases of the personnel of the institutions does not facilitate being able to deepen the investigation.

4.6 Implications for HEI

In higher education, knowledge is essential to generate competitive advantages. Therefore, HEIs must know the implications of developing knowledge management processes with intellectual capital and innovation. Therefore, to improve the performance, prestige, and impact of HEIs, it is necessary to establish strategies, policies, and culture to develop the above factors, in which all academic and administrative staff actively participate.

4.7 Future lines of research

As a future line of research, it is proposed to study the three variables analyzed from the perspective of the different disciplines within HEIs and their relationship with their performance. In this case, the contribution by discipline is the same as the institution's overall performance? For example, do intellectual capital and knowledge management impact the same proportion on performance according to the type of discipline?

Data availability Data available on request from the authors.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

References

- Adeinat, I. M., & Abdulfatah, F. H. (2019). Organizational culture and knowledge management processes: case study in a public university. *VINE Journal of Information and Knowledge Management Systems*, 49(1), 35–53. <https://doi.org/10.1108/vjikms-05-2018-0041>
- Agostini, L., Nosella, A., & Filippini, R. (2017). Does intellectual capital allow improving innovation performance? A quantitative analysis in the SME context. *Journal of Intellectual Capital*, 18(2), 400–418. <https://doi.org/10.1108/JIC-05-2016-0056>
- Akil, S. R., Soemaryani, I., Hilmiana, H., & Joeliaty, J. (2021). Determinant factors of intellectual capital for improving public sector innovation: an empirical study from Indonesia. *The Journal of Asian Finance Economics and Business*, 8(12), 421–429. <https://doi.org/10.13106/JAFEB.2021.VOL8.NO12.0421>
- Allameh, S. M. (2018). Antecedents and consequences of intellectual capital. *Journal of Intellectual Capital*, 19(1), 858–874. <https://doi.org/10.1108/jic-05-2017-0068>

- Alshamsi, S., Isaac, O., & Bhaumik, A. (2019). The effects of intellectual capital on organizational innovation within Abu Dhabi police in UAE. *International Journal on Emerging Technologies*, 10(1), 50–58.
- Al-Kurdi, O., El-Haddadeh, R., & Eldabi, T. (2018). Knowledge sharing in higher education institutions: a systematic review. *Journal of Enterprise Information Management*, 31(2), 226–246. <https://doi.org/10.1108/jeim-09-2017-0129>
- Andersen, A. (1999). *Modelo de la Gestión del Conocimiento de Arthur Andersen*. Madrid, Spain. Retrieved March 27, 2020, from www.madrimasd.org/revista/revista28aula/aula3.asp
- Ardichvili, A., Maurer, M., Li, W., Wentling, T., & Stuedemann, R. (2006). Cultural influences on knowledge sharing through online communities of practice. *Journal of Knowledge Management*, 10(1), 94–107. <https://doi.org/10.1108/13673270610650139>
- Barley, W. C., Treem, J. W., & Kuhn, T. (2018). Valuing multiple trajectories of knowledge: a critical review and agenda for knowledge management research. *Academy of Management Annals*, 12(1), 278–317. <https://doi.org/10.5465/annals.2016.0041>
- Bercovitz, J., & Feldmann, M. (2006). Entrepreneurial universities and technology transfer: a conceptual framework for understanding knowledge-based economic development. *Journal of Technology Transfer*, 31(1), 175–188. <https://doi.org/10.1007/s10961-005-5029-z>
- Bontis, N., Bart, C., Ramírez, Y., Lorduy, C., & Rojas, J. (2007). Intellectual capital management in spanish universities. *Journal of Intellectual Capital*, 8(4), 732–748. <https://doi.org/10.1108/14691930710830873>
- Booker, L. D., Bontis, N., & Serenko, A. (2008). The relevance of knowledge management and intellectual capital research. *Knowl Process Mgmt*, 15(99), 235–246. <https://doi.org/10.1002/kpm.314>
- Cabrilo, S., & Dahms, S. (2018). How strategic knowledge management drives intellectual capital to superior innovation and market performance. *Journal of Knowledge Management*, 22(3), 621–648. <https://doi.org/10.1108/JKM-07-2017-0309>
- Carmines, E. G., & Zeller, R. A. (1979). *Reliability and validity assessment* (17 vol.). Sage Publications.
- Chawla, D., & Joshi, H. (2011). Impact of knowledge management on learning organization in Indian organizations—A comparison. *Know Process Mgmt*, 18(1), 266–277. <https://doi.org/10.1002/kpm.384>
- Cheng, M. Y., Ho, J. S. Y., & Lau, P. M. (2009). Knowledge sharing in academic institutions: a study of multimedia university Malaysia. *Electronic Journal of Knowledge Management*, 7(3), 313–324.
- Chin, W. W. (1998). The partial least squares approach to structural equation modeling. *Modern Methods for Business Research*, 295(2), 295–336.
- Cho, T., & Korte, R. (2014). Managing knowledge performance: testing the components of a knowledge management system on organizational performance. *Asia Pacific Education Review*, 15(2), 313–327. <https://doi.org/10.1007/s12564-014-9333-x>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates, Publishers.
- Cricelli, L., Greco, M., Grimaldi, M., & Llanes Dueñas, L. (2018). Intellectual capital and university performance in emerging countries. *Journal of Intellectual Capital*, 19(1), 71–95. <https://doi.org/10.1108/JIC-02-2017-0037>
- Damanpour, F., Walker, R., & Avellaneda, C. (2009). Combinative effects of innovation types and organizational performance: a longitudinal study of service organizations. *Journal of Management Studies*, 46(4), 650–675. <https://doi.org/10.1111/j.1467-6486.2008.00814.x>
- Davenport, T. H. (1994). Saving its soul: human-centered information management. *Harvard Business Review*, 72(2), 119–131.
- Davenport, T., & Hansen, M. (1999). *Knowledge management at Andersen Consulting*. Harvard Business School Case Study, 9-499-032. Harvard Business Press.
- De Frutos-Belizón, J., Martín-Alcázar, F., & Sánchez-Gardey, G. (2019). Conceptualizing academic intellectual capital: definition and proposal of a measurement scale. *Journal of Intellectual Capital*, 20(3), 306–334. <https://doi.org/10.1108/JIC-09-2018-0152>
- De Matos, P. E., Alves, H., & Leitão, J. (2022). In search of intangible connections: intellectual capital, performance and quality of life in higher education institutions. *High Education*, 83, 243–260. <https://doi.org/10.1007/s10734-020-00653-9>
- De Matos, P. E., Leitão, J., & Alves, H. (2020). Bridging intellectual capital, sustainable development and quality of life in higher education institutions. *Sustainability*, 12(2), 479. <https://doi.org/10.3390/su12020479>

- Demuner, M., Nava, R., & Ibarra, M. (2016). Dimensiones e indicadores de capital estructural para la universidad pública. In Nava, R., Mercado, P., & Demuner, M. (Eds.), *El capital intelectual en la universidad pública*, (pp. 61–80). Universidad Autónoma del Estado de México.
- Dias Jordão, R., & Novas, C. (2017). Knowledge management and intellectual capital in networks of small- and medium-sized enterprises. *Journal of Intellectual Capital*, 18(3), 667–692. <https://doi.org/10.1108/JIC-11-2016-0120>
- Ding, W., Choi, E., & Aoyama, A. (2019). Relational study of wise (phronetic) leadership, knowledge management capability, and innovation performance. *Asia Pacific Management Review*, 24(4), 310–317. <https://doi.org/10.1016/j.apmr.2018.10.005>
- Donate, M. J., & de Sánchez, J. D. (2015). The role of knowledge-oriented leadership in knowledge management practices and innovation. *Journal of Business Research*, 68(2), 360–370. <https://doi.org/10.1016/j.jbusres.2014.06.022>
- Duhon, B. (1998). It's all in our heads. *Inform*, 21(8), 8–14.
- Dzenopoljac, V., Alasadi, R., Zaim, H., & Bontis, N. (2018). Impact of knowledge management processes on business performance: evidence from Kuwait. *Knowledge and Process Management*, 25(1), 77–87. <https://doi.org/10.1002/kpm.1562>
- Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2009). G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>
- Ferasso, M., & Grenier, C. (2021). Fostering SME's co-development of innovative projects in biotech clusters: extending the sets of enablers for the knowledge creation process. *Technology in Society*, 67, 101729. <https://doi.org/10.1016/j.techsoc.2021.101729>
- Fernández-López, S., Rodeiro-Pazos, D., Calvo, N., & Rodríguez-Gulías, M. (2018). The effect of strategic knowledge management on the universities' performance: an empirical approach. *Journal of Knowledge Management*, 22(3), 567–586. <https://doi.org/10.1108/JKM-08-2017-0376>
- Fornell, C., & Larcker, D. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.1177/002224378101800104>
- Gold, A. H., Malhotra, A., & Segars, A. H. (2001). Knowledge management: an organizational capabilities perspective. *Journal of Management Information Systems*, 18(1), 185–214. <https://doi.org/10.1080/07421222.2001.11045669>
- Haider, S. A., Akbar, A., Tehseen, S., Poulava, P., & Jaleel, F. (2022). The impact of responsible leadership on knowledge sharing behavior through the mediating role of person–organization fit and moderating role of higher educational institute culture. *Journal of Innovation & Knowledge*, 7(4), 100265. <https://doi.org/10.1016/j.jik.2022.100265>
- Hair, J., Sarstedt, M., Hopkins, L., & Kuppelwieser, V. (2014). Partial least squares structural equation modeling (PLS-SEM): an emerging tool in business research. *European Business Review*, 26(2), 106–121. <https://doi.org/10.1108/EBR-10-2013-0128>
- Hayaean, S., Hesarzadeh, R., & Abbaszadeh, M. R. (2022). The impact of knowledge management strategies on the relationship between intellectual capital and innovation: evidence from SMEs. *Journal of Intellectual Capital*, 23(4), 765–798. <https://doi.org/10.1108/JIC-07-2020-0240>
- Henseler, J., & Dijkstra, T. (2015). In C. Modeling (Ed.), *ADANCO 2.1.1*. Kleve.
- Henseler, J., Ringle, C., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Huang, J. W., & Li, Y. H. (2009). The mediating effect of knowledge management on social interaction and innovation performance. *International Journal of Manpower*, 30(3), 285–301. <https://doi.org/10.1108/01437720910956772>
- Iacoviello, G., Bruno, E., & Cappiello, A. (2019). A theoretical framework for managing intellectual capital in higher education. *International Journal of Educational Management*, 33(5), 919–938. <https://doi.org/10.1108/IJEM-02-2018-0080>
- Iqbal, A. (2021). Innovation speed and quality in higher education institutions: the role of knowledge management enablers and knowledge sharing process. *Journal of Knowledge Management*, 25(9), 2334–2360. <https://doi.org/10.1108/JKM-07-2020-0546>
- Iqbal, A., Latif, F., Marimon, F., Sahibzada, U., & Hussain, S. (2019). From knowledge management to organizational performance: modelling the mediating role of innovation and intellectual capital in higher education. *Journal of Enterprise Information Management*, 32(1), 36–59. <https://doi.org/10.1108/JEIM-04-2018-0083>

- Kline, R. B. (2015). *Principles and practice of structural equation modeling*. Guilford Publication.
- Leitner, K. H., Curaj, A., Elena-Perez, S., Fazlagic, J., Kalemis, K., Martinaitis, Z., Secundo, G., Sicilia, M. A., & Zaks, K. (2014). *A strategic approach for intellectual capital management in European universities*. Guidelines for implementation, UEFISCDI Blueprint Series No. 1, Executive Agency for Higher Education, Research, Development and Innovation Funding, Bucharest. Retrieved March 18, 2021, from <https://psaar.net/abstracts/231-a-strategic-approach-for-intellectual-capital-management-in-european-universities-guidelines-for-implementation>
- Lin, H. F. (2007). Knowledge sharing and firm innovation capability: an empirical study. *International Journal of Manpower*, 28(3–4), 315–332. <https://doi.org/10.1108/01437720710755272>
- Ling, T. C., & Nasurdin, A. Z. (2010). The influence of knowledge management effectiveness on administrative innovation among Malaysian manufacturing firms. *ASIA Academy of Management Journal*, 15(1), 63–67.
- Martin-Sardesai, A., & Guthrie, J. (2018). Human capital loss in an academic performance measurement system. *Journal of Intellectual Capital*, 19(1), 53–70. <https://doi.org/10.1108/JIC-06-2017-0085>
- Metha, A. M., & Tariq, M. (2020). An institution based view towards innovation strategy and knowledge management in the European Region. *Journal of Management Information and Decision Sciences*, 23(4), 223–228.
- Mubarak, K., & Sabraz Nawaz, S. (2019). Knowledge management strategic enablers in higher education institutions in Sri Lanka. *International Journal of Grid and Distributed Computing*, 12(3), 190–198.
- Nonaka, I., & Takeuchi, H. (1995). *The knowledge creating company*. Oxford University Press.
- Obeidat, B., Al-Suradi, M., Masa'deh, R., & Tarhini, A. (2016). The impact of knowledge management on innovation: an empirical study on Jordanian consultancy firms. *Management Research Review*, 39(10), 1214–1238. <https://doi.org/10.1108/MRR-09-2015-0214>
- Paoloni, P., Cesaroni, F. M., & Demartini, P. (2019). Relational capital and knowledge transfer in universities. *Business Process Management Journal*, 25(1), 185–201. <https://doi.org/10.1108/BPMJ-06-2017-0155>
- Pedro, E., Leitão, J., & Alves, H. (2019). The intellectual capital of higher education institutions: operationalizing measurement through a strategic prospective lens. *Journal of Intellectual Capital*, 20(3), 355–381. <https://doi.org/10.1108/JIC-07-2018-0117>
- Pepple, D., Makama, C., & Okeke, J. (2022). Knowledge management practices: a public sector perspective. *Journal of Business Research*, 153, 509–516. <https://doi.org/10.1016/j.jbusres.2022.08.041>
- Pflugfelder, N. S. (2021). Knowledge management as a driver of performance in ambulatory healthcare – a systematic literature review through an intellectual capital lens. *Journal of Intellectual Capital*, 22(2), 403–432. <https://doi.org/10.1108/JIC-02-2020-0068>
- Pinho, I., Rego, A., Pina, A., & Cunha, M. (2012). Improving knowledge management processes: a hybrid positive approach. *Journal of Knowledge Management*, 16(2), 215–242. <https://doi.org/10.1108/13673271211218834>
- Quarchioni, S., Paternostro, S., & Trovarelli, F. (2022). Knowledge management in higher education: a literature review and further research avenues. *Knowledge Management Research & Practice*, 20(2), 304–319. <https://doi.org/10.1080/14778238.2020.1730717>
- Qurashi, I. A., Khalique, M., Ramayah, T., Bontis, N., & Yaacob, M. R. (2020). Impact of intellectual capital on innovation in pharmaceutical manufacturing SMEs in Pakistan. *International Journal of Learning and Intellectual Capital*, 17(1), 61–76. <https://doi.org/10.1504/IJLIC.2020.105324>
- Raj Adhikari, D. (2010). Knowledge management in academic institutions. *International Journal of Educational Management*, 24(2), 94–104. <https://doi.org/10.1108/09513541011020918>
- Ramírez Córcoles, Y., Peñalver, S., & Tejada, Á. (2011). Intellectual capital in Spanish public universities: stakeholders' information needs. *Journal of Intellectual Capital*, 12(3), 356–376. <https://doi.org/10.1108/14691931111154689>
- Ramírez, Y., & Gordillo, S. (2014). Recognition and measurement of intellectual capital in Spanish universities. *Journal of Intellectual Capital*, 15(1), 173–188. <https://doi.org/10.1108/JIC-05-2013-005>
- Ramírez, Y., & Tejada, Á. (2019). Digital transparency and public accountability in Spanish universities in online media. *Journal of Intellectual Capital*, 20(5), 701–732. <https://doi.org/10.1108/JIC-02-2019-0039>
- Ramírez, Y., Santos, J., & Tejada, A. (2012). Demanda de información sobre capital intelectual en las universidades públicas españolas. *Cuadernos de Gestión*, 12(1), 83–106.
- Raudeliūniene, J., & Kordab, M. (2019). Impact of knowledge oriented leadership on knowledge management processes in the middle eastern audit and consulting companies. *Business Management and Education*, 17(1), 248–268. <https://doi.org/10.3846/bme.2019.11284>

- Rehman, U. U., & Iqbal, A. (2020). Nexus of knowledge-oriented leadership, knowledge management, innovation and organizational performance in higher education. *Business Process Management Journal*, 26(6), 1731–1758. <https://doi.org/10.1108/BPMJ-07-2019-0274>
- Rezaei, R., & Moghanlo, F. (2015). The effect of intellectual capital on organizational innovation in the agricultural services ventures of Zanjan province. *Journal of Entrepreneurship Development*, 7(4), 653–673. <https://doi.org/10.22059/JED.2014.53623>
- Ringle, C., Sarstedt, M., Mitchell, R., & Gudergan, S. (2018). Partial least squares structural equation modeling in HRM research. *The International Journal of Human Resource Management*, 31(12), 1617–1643. <https://doi.org/10.1080/09585192.2017.1416655>
- Rivera, G., & Rivera, I. (2016). Diseño, medición y análisis de un modelo para la gestión del conocimiento en el contexto de una universidad mexicana. *innovar*, 26(59), 21–34. <https://doi.org/10.15446/innovar.v26n59.54320>
- Sahibzada, U. F., Jianfeng, C., Latif, K. F., & Sahibzada, H. F. (2020a). Knowledge management processes, knowledge worker satisfaction, and organizational performance: symmetric and asymmetrical analysis. *Aslib Journal of Information Management*, 72(1), 112–129.
- Sahibzada, U. F., Jianfeng, C., Latif, K. F., & Sahibzada, H. F. (2020b). Fueling knowledge management processes in Chinese higher education institutes (HEIs): the neglected mediating role of knowledge worker satisfaction. *Journal of Enterprise Information Management*, 33(6), 1395–1417. <https://doi.org/10.1108/JEIM-07-2019-0197>
- Sahibzada, U. F., Latif, K. F., & Xu, Y. (2022). Symmetric and asymmetric modeling of knowledge management enablers to knowledge management processes and knowledge worker productivity in higher education institutes. *Journal of Enterprise Information Management*, 35(3), 729–756.
- Sadeghi Boroujerdi, S., Hasani, K., & Delshab, V. (2019). Investigating the influence of knowledge management on organizational innovation in higher educational institutions. *Kybernetes*, 49(2), 442. <https://doi.org/10.1108/K-09-2018-0492>
- Secretaría de Educación Pública (2018, November 05). *Subsecretaría de Planeación, Evaluación y Coordinación. Dirección General de Planeación, Programación y Estadística Educativa*. Retrieved May 5, 2020, from https://spec.sep.gob.mx/web/?page_id=28
- Secundo, G., Dumay, J., Schiuma, G., & Passiante, G. (2016). Managing intellectual capital through a collective intelligence approach. *Journal of Intellectual Capital*, 17(2), 298–319. <https://doi.org/10.1108/JIC-05-2015-0046>
- Secundo, G., Lombardi, R., & Dumay, J. (2018). Intellectual capital in education. *Journal of Intellectual Capital*, 19(1), 2–9. <https://doi.org/10.1108/JIC-10-2017-0140>
- Shariq, S. M., Mukhtar, U., & Anwar, S. (2019). Mediating and moderating impact of goal orientation and emotional intelligence on the relationship of knowledge-oriented leadership and knowledge sharing. *Journal of Knowledge Management*, 23(2), 332–350. <https://doi.org/10.1108/JKM-01-2018-0033>
- Sultanova, G., Svyatov, S., & Ussenbayev, N. (2018). Transmitting competencies at universities in Kazakhstan: intellectual capital of teachers. *Journal of Intellectual Capital*, 19(1), 112–134. <https://doi.org/10.1108/JIC-04-2017-0058>
- Supapawawisit, B., Chandrachai, A., & Thawesaengskulthai, N. (2018). The critical factors of research and innovation creation in public universities in Thailand. *International Journal of Trade and Global Markets*, 12(1–2), 109–111.
- Tehseen, S., Ramayah, T., & Sajilan, S. (2017). Testing and controlling for common method variance: a review of available methods. *Journal of Management Sciences*, 4(2), 142–168. <https://doi.org/10.20547/jms.2014.1704202>
- Teixeira, E. K., Oliveira, M., & Curado, C. M. M. (2018). Knowledge management process arrangements and their impact on innovation. *Business Information Review*, 35(1), 29–38. <https://doi.org/10.1177/0266382118757771>
- Tether, B. (2003). The sources and aims of innovation in services: variety between and within sectors. *Economics of Innovation and New Technology*, 12(6), 481–505. <https://doi.org/10.1080/104385902200029221>
- Thani, F. N., & Mirkamali, S. M. (2018). Factors that enable knowledge creation in higher education: a structural model. *Data Technologies and Applications*, 52(3), 424–444. <https://doi.org/10.1108/dta-10-2016-0068>
- Torres, A., Santos-Ferraz, S., & Santos-Rodrigues, H. (2018). The impact of knowledge management factors in organizational sustainable competitive advantage. *Journal of Intellectual Capital*, 19(2), 453–472. <https://doi.org/10.1108/JIC-12-2016-0143>

- Veer Ramjeawon, P., & Rowley, J. (2020). Enablers and barriers to knowledge management in universities: perspectives from South Africa and Mauritius. *Aslib Journal of Information Management*, 72(5), 745–764. <https://doi.org/10.1108/AJIM-12-2019-0362>
- Wang, Z., Cai, S., Liang, H., Wang, N., & Xiang, E. (2021). Intellectual capital and firm performance: the mediating role of innovation speed and quality. *The International Journal of Human Resource Management*, 32(6), 1222–1250. <https://doi.org/10.1080/09585192.2018.1511611>
- Wang, Z., & Wang, N. (2012). Knowledge sharing, innovation and firm performance. *Expert Systems with Applications*, 39(10), 8899–8908. <https://doi.org/10.1016/j.eswa.2012.02.017>
- Wang, Z., Wang, N., Cao, J., & Ye, X. (2016). The impact of intellectual capital – knowledge management strategy fit on firm performance. *Management Decision*, 54(8), 1861–1885. <https://doi.org/10.1108/MD-06-2015-0231>
- Wang, Z., Wang, N., & Liang, H. (2014). Knowledge sharing, intellectual capital and firm performance. *Management Decision*, 52(2), 230–258. <https://doi.org/10.1108/MD-02-2013-0064>
- Waseem, B., Loo-See, B., Adeel, A., & Riaz, A. (2018). Impact of intellectual capital on innovation capability and organizational performance: an empirical investigation. *Serbian Journal of Management*, 13(1), 365–379. <https://doi.org/10.5937/sjm13-16997>
- Wen Chong, C., Holden, T., Wilhelmij, P., & Schmidt, R. A. (2000). Where does knowledge management add value? *Journal of Intellectual Capital*, 1(4), 366–380. <https://doi.org/10.1108/14691930010359261>
- Wendra, W., Sule, E. T., Joeliaty, J., & Azis, Y. (2019). Exploring dynamic capabilities, intellectual capital and innovation performance relationship: evidence from the garment manufacturing. *Business: Theory and Practice*, 20(1), 123–136. <https://doi.org/10.3846/btp.2019.12>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.